Fiscal Year:	FY 2013	**Task Last Updated:**	FY 07/05/2012
PI Name:	Shay, Jerry W. Ph.D.		
Project Title:	Risk Assessment of Space Radiation-Enhanced Colon Tumorigenesis		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jerry.shay@utsouthwestern.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	214-648-3282
Organization Name:	University of Texas Southwestern Medical Center		
PI Address 1:	Cell Biology Department		
PI Address 2:	5323 Harry Hines Blvd		
PI Web Page:			
City:	Dallas	**State:**	TX
Zip Code:	75390-7208	**Congressional District:**	30
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Space Radiobiology NNJ08ZSA001N
Start Date:	10/01/2008	**End Date:**	10/01/2012
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
Contact Email:	lisa.c.simonsen@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 10/1/2012 to accommodate final reporting (Ed., 7/9/2012) NOTE: NCE to 9/30/2012 per C. Guidry/JSC and NSSC information (Ed., 9/27/2011)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX08BA54G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This proposal addresses the Carcinogenesis category of the Bioastronautics Roadmap by investigating carcinogenic risk of human colonic cells and tissues after low and high LET radiation exposure. The colon is a major target for the carcinogenic effects of radiation. Aberrant crypt foci are an early precursor lesion for colorectal cancer (CRC) that is characterized by trisomy 7 (present in 37% of adenomas and 80% of adenocarcinomas). The appearance of the adenomatous polyp becomes more prevalent with increased age (10% incidence at age 40 and >25% by age 60). Pre-cancerous lesions may already exist or develop in astronauts going on long-term missions, the result of which would be an increased risk for cancer development through exposure to space radiation. Our key objective is to determine the probabilities for increased colonic carcinogenesis from space radiation as a result of radiation dose, quality, and dose rate. We will expose 2D and 3D human colonic cells and human colonic biopsies to energetic protons (H), high Z and energy particles (HZE), and x-rays and analyze results using both biological assays and genome-wide systems approaches. We are comparing these results to a novel immortalized human trisomy 7 colonic epithelial cell line and will model carcinogenic risk of these cells compared to normal and immortalized diploid colonic cells. We have also initiated a small molecule screen to identify non toxic compounds that may inhibit the trisomy 7 cells but not the normal diploid. We has also examined DNA damage sensing, signaling, and repair kinetics in normal and trisomy 7 human colonic cell lines. Finally, we have obtained a substantial amount of information about a novel oral bioavailable compound CDDO-Me (bardoxolone methyl, RTA 402) and have determined it is an effective and non toxic countermeasure for protecting normal colonic cells from the detrimental effects of low and high LET irradiation. As part of a new project we are extending these countermeasure studies to mouse models. This research should improve our mechanistic understanding for the variability of radiation dose, quality, and dose-rate dependencies for colon cancer development and provide biological markers for modeling and risk assessment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The development of methods to maintain human colonic epithelial cells in culture long term has many benefits for life on Earth. The unique human colonocyte reagents we have developed may help in our understanding of the earliest events in the development of human colon cancer. In addition to normal human colonocytes, we have also isolated the first pre-cancerous human colonocyte cell culture that has one extra chromosome (trisomy 7). Trisomy 7 occurs in a large fraction of patients with sporadic colon cancer and is believed to be one of the earliest steps in colon cancer progression. Our goal is to identify the underlying molecular mechanisms leading to early genetic changes in colon cancer and to validate novel genetic alterations that may occur as critical steps in colon cancer progression. We have initiated experiments to silence and express candidate genes known to be altered in colon cancer into our colonocyte cell lines. Experiments conducting genome-wide sequencing of colon tumors revealed that in addition to the common already known gene mutations, many additional genes are mutated in colon cancer. It is not known which of these mutated genes are important for colon cancer progression. Development of an increasingly cancerous phenotype in the colonocytes after specific genetic alterations should help validate the currently known pathways leading to colon cancer and identify new targets for therapeutic interventions. This work has mostly been accomplished and additional studies are required to complete them. We are also investigating a novel biological countermeasure that may protect against radiation damage. This would also have a major impact on cancer patients being treated with therapeutic radiation.
Task Progress:	1. DNA damage in human colonic epithelial cells is more difficult to repair when caused by iron particles compared to protons and gamma rays. 2. Iron damage persists longer in both 2D and 3D human colonic epithelial cell cultures compared to protons and gamma rays. 3. Epithelial cells in intact human colonic tissues display more damage foci than surrounding stromal cells after both high and low LET radiations. All cells in intact colonic tissues are exposed to the same radiation dosage and are likely experiencing equal amounts of DNA double strand breaks, but location in the cell cycle may explain why the replicating epithelial cells show more pronounced DNA damage response compared to quiescent stromal cells. 4. Additional cells from another patient have been established and characterized. We have an immortalized normal diploid cell line and one that is immortalized with one chromosomal change (trisomy 7). Trisomy 7 is present in about 65% of all colonic adenoma and adenocarcinomas. We have now progressed these cells partially toward tumorigenicity by reducing p53 expression (shRNA to p53), overexpressing K-ras and c- myc and expressing a truncated APC. The characterization of these cells and strategies to identify important tumor suppressors have been published (see end of the report). We continue to test if irradiation converts these cells to cancer forming cells so risk assessment can be modeled. 5. We have now demonstrated and published (Gastroenterology, 138:1012-1021, 2010) the ability to differentiate the immortalized human colonic epithelial cell lines We showed that our human colonic epithelial cells express several gastrointestinal putative stem cell markers including CD44, CD29, BMI-1, and LGR5. This indicates that our normal diploid hTERT immortalized human colonic epithelial cells can make similar 3D differentiated structures as do primary cells from mouse small intestine. Our cells are clonal, can divide long-term (are immortalized), and single cells divide to generate self-organizing structures with several differentiated cell types and architecture of crypts present in adults in the absence of a cellular niche. Our cells demonstrate polarized villin staining, junction formation (ZO-1), a subset of cells express Muc2 similarly to the staining of mouse small intestinal cells, and some even express chromogranin A (a neuroendocrine marker). This also helps demonstrate that we are culturing human colonic epithelial stem cells. 6. We have also passaged the cells with trisomy 7 (contain hTERT, cdk4, shRNAp53 and mutant K-ras) and observed that a subclone of these cells contain a trisomy 20 (also very common in human colon cancer). While these cells make large colonies and grow anchorage independent in soft agar they do not make tumors in nude mice. We have now expressed c-myc in these cells and these also do not make tumors. We have also introduced a key mutant version of APC into these cells. These isogenic partially progressed cell lines are extremely useful for radiation experiments to determine biological steps in cancer progression. Importantly, our results demonstrate that it requires multiple independent changes in a single cell in order to progress a normal human colonic epithelial cells into a fully malignant (lethal) cancer cell. This suggests that the exposure to the type and doses likely to be experienced by astronauts on a long-term mission into space are less likely to have a major impact on lethal cancer formation. However, more research needs to be done to confirm this speculation. 7. Biological countermeasure experiments. We have conducted a significant number of experiments with methyl-2-cyano-3,12-dioxooleana-1,9(11)-dien-28-oate (BARD, also termed RTA-402, or CDDO-Me). BARD is an Nrf2 mediated phase II enzyme activator with a well understood molecular mechanism of action. Importantly, BARD is

	a completely synthetic compound that is non toxic and orally available. In two completed Phase II clinical trials BARD taken orally once-daily not only mimics the body's natural mechanism for resolving inflammation but also reduces the risk of death in patients with stage 3 chronic kidney disease. While anti-oxidant inflammation modulators (AIMs) such as BARD would be expected to provide risk reduction for low-LET radiation, it is much less certain if AIMs would be effective in protecting against high-LET irradiation. In our research, we have documented that BARD is an effective biological countermeasure by reducing high-LET irradiation associated human epithelial cell transformation.
Bibliography Type:	Description: (Last Updated: 11/21/2025)
Abstracts for Journals and Proceedings	Kaisani A, Kim SB, Delgado O, Richardson JA, Wright WE, Story MD, Minna JD, Shay JW. "Risk Assesment of Radiation Induced Invasive Cancer in Mouse Models of Lung and Colon Cancers." Presented as a poster; student poster contest. 22nd Annual NASA Space Radiation Investigators' Workshop, League City, Texas, September 18–21, 2011. 22nd Annual NASA Space Radiation Investigators' Workshop, League City, Texas, September 18–21, 2011. http://www.dsls.usra.edu/meetings/radiation2011/pdf/7120.pdf , Sep-2011
Abstracts for Journals and Proceedings	Kim SB, Eskiocak U, Ly P, Kaisani A, Biglione S, Shay JW. "Bardoxylone methyl (CDDO-Me): an antioxidant, anti-inflammatory modulator is a novel radiation countermeasure and mitigator. " Presented as a postdoctoral poster contest. 22nd Annual NASA Space Radiation Investigators' Workshop, League City, Texas, September 18–21, 2011. 22nd Annual NASA Space Radiation Investigators' Workshop, League City, Texas, September 18–21, 2011. http://www.dsls.usra.edu/meetings/radiation2011/pdf/7030.pdf , Sep-2011
Articles in Peer-reviewed Journals	Kim SB, Pandita RK, Eskiocak U, Ly P, Kaisani A, Kumar R, Cornelius C, Wright WE, Pandita TK, Shay JW. "Targeting of Nrf2 enhances DNA doublestrand break repair and protects colonic epithelial cell after irradiation." PNAS (accepted pending minor revisions, 2012). , Jul-2012
Articles in Peer-reviewed Journals	Zhang L, Komurov K, Wright WE, Shay JW. "Identification of novel driver tumor suppressors through functional interrogation of putative passenger mutations in colorectal cancer." International Journal of Cancer, 2012 Jun 30. [Epub ahead of print] PMID: 22753261 ; http://dx.doi.org/10.1002/ijc.27705 , Jun-2012
Articles in Peer-reviewed Journals	Ly P, Kim SB, Wright WE, Shay JW. "AICAR selectively inhibits trisomy 7 human colonic epithelial cell proliferation through EGFR degradation." Oncogene, (in press, 2012). , Jul-2012